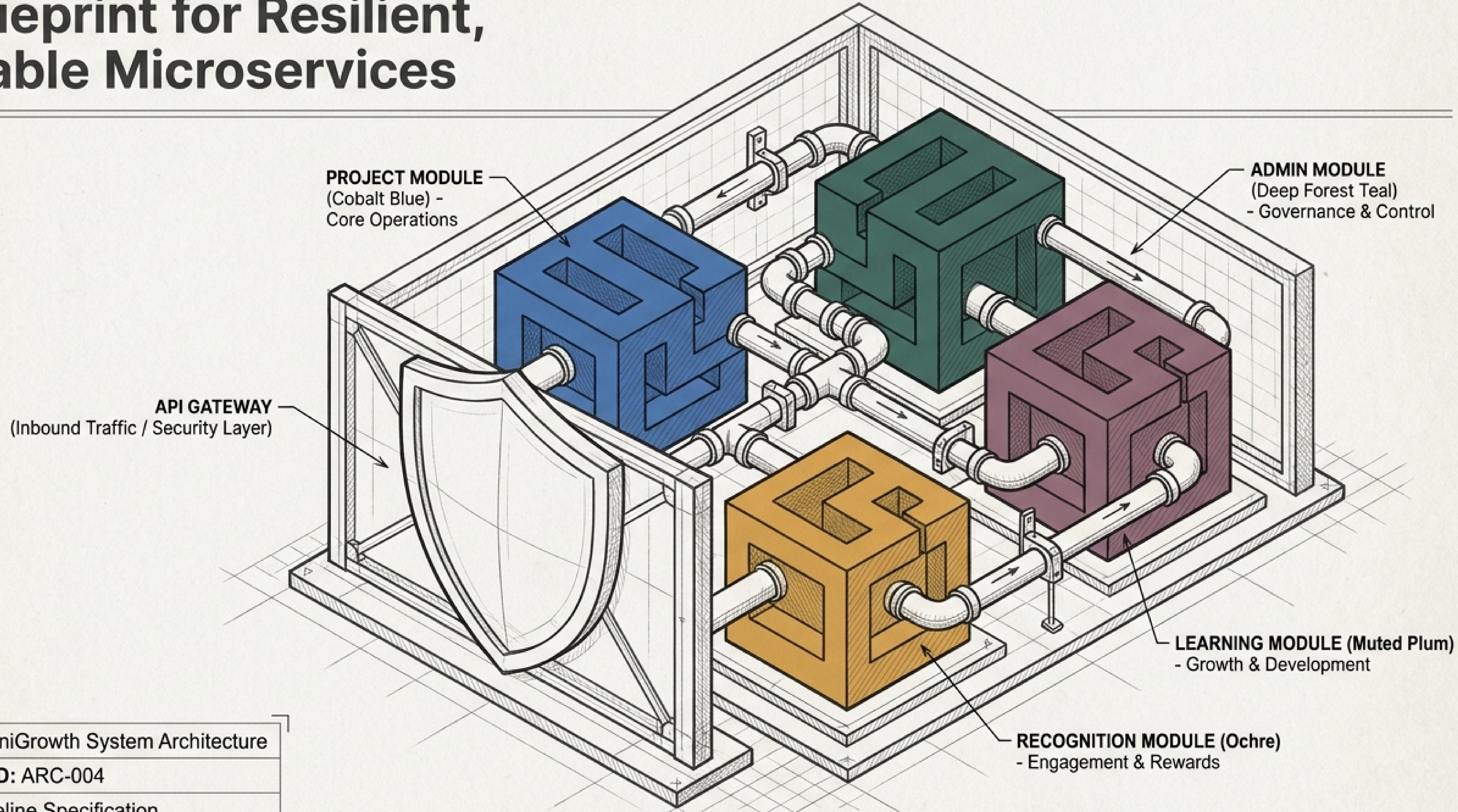
A Blueprint for Resilient, Scalable Microservices



Project: OmniGrowth System Architecture	
Document ID: ARC-004	
Status: Baseline Specification	

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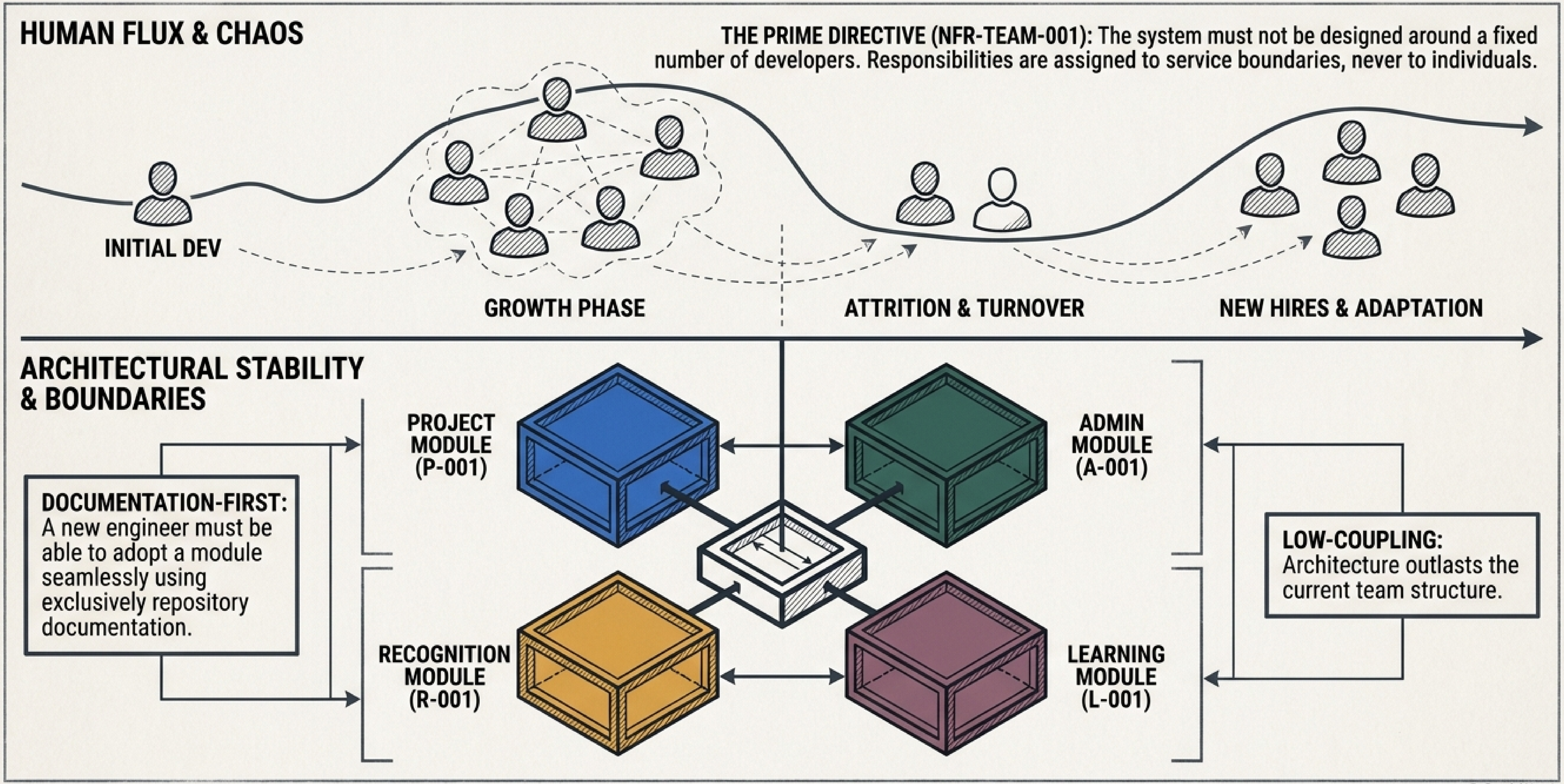
ORGANISATIONAL SCALE MUST NOT DICTATE SYSTEM ARCHITECTURE



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FOUR STRICTLY BOUNDED BUSINESS DOMAINS FORM THE CORE ENGINE

QUADRANT 1: PROJECT & PROCESS ORCHESTRATION

Domain Ownership: Projects, tasks, workflow states, and dependencies.

Key Output: Durable completion tracking and state traceability.

QUADRANT 2: ADMINISTRATIVE HUB

Domain Ownership: User profiles, roles, and granular permissions.

Key Output: Centralised identity and RBAC context for all services.

QUADRANT 3: RECOGNITION & INCENTIVES

Domain Ownership: Achievement records, point ledgers, and rewards.

Key Output: Idempotent point awards and duplicate-processing protection.

QUADRANT 4: HOLISTIC LEARNING FRAMEWORK

Domain Ownership: Professional development dimensions and enrolments.

Key Output: Progress tracking and controlled AI integration points.

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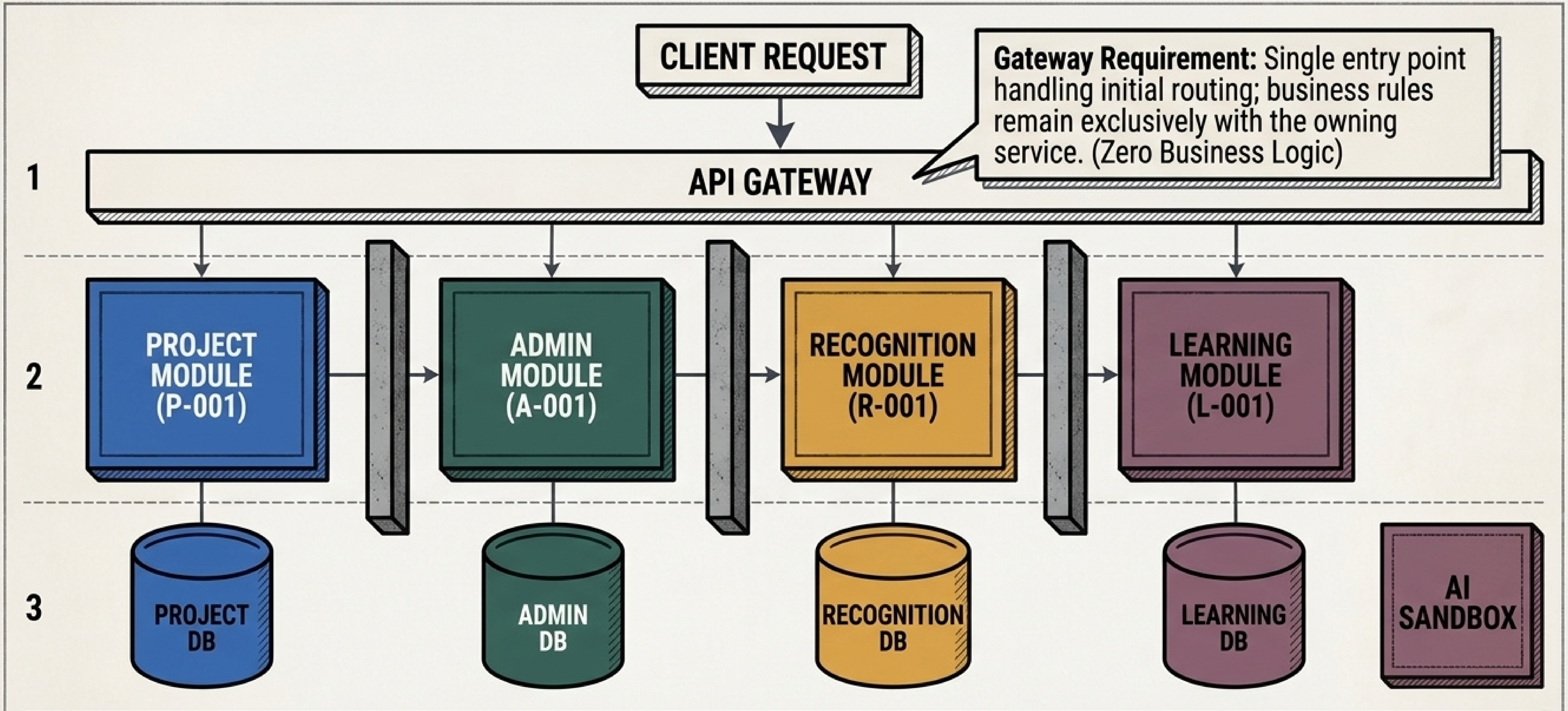
AN IMPENETRABLE BOUNDARY SEPARATES CORE SERVICES FROM EDGE ROUTING



END-TO-END INTEGRATION: Services communicate solely through documented APIs and structured events.

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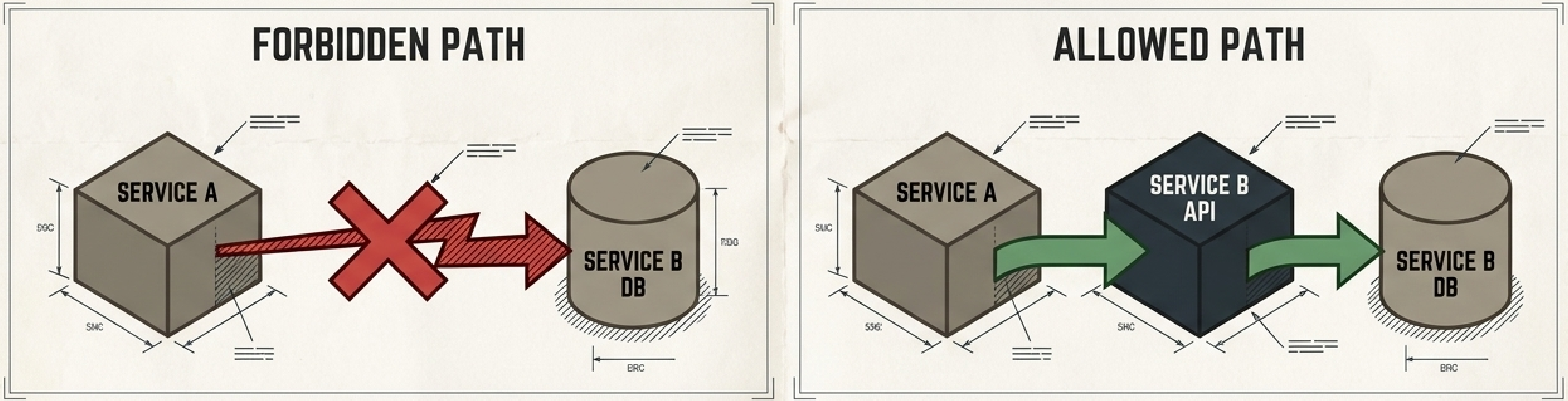
ABSOLUTE DATA ISOLATION GUARANTEES FAILURE CONTAINMENT

RULE NFR-DAT-001: NO DIRECT CROSS-SERVICE DATABASE READS OR WRITES.



THE RATIONALE: SHARED DATABASES CREATE HIDDEN COUPLING. STRICT DATA OWNERSHIP ENSURES CLEAR SERVICE BOUNDARIES AND PREVENTS CASCADING DATABASE LOCKS. EXTERNAL REFERENCES MUST USE STABLE IDS, NOT FOREIGN KEYS.

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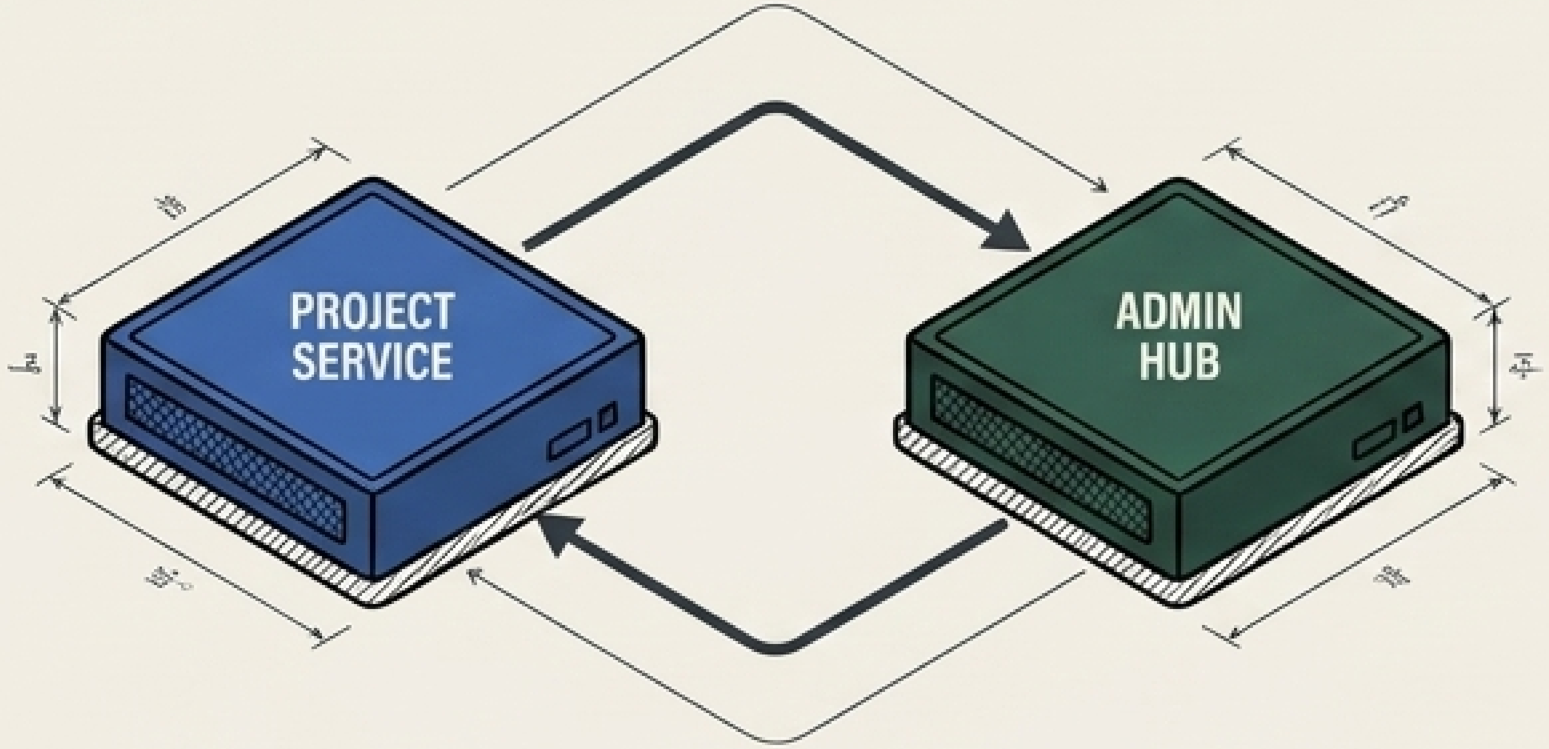
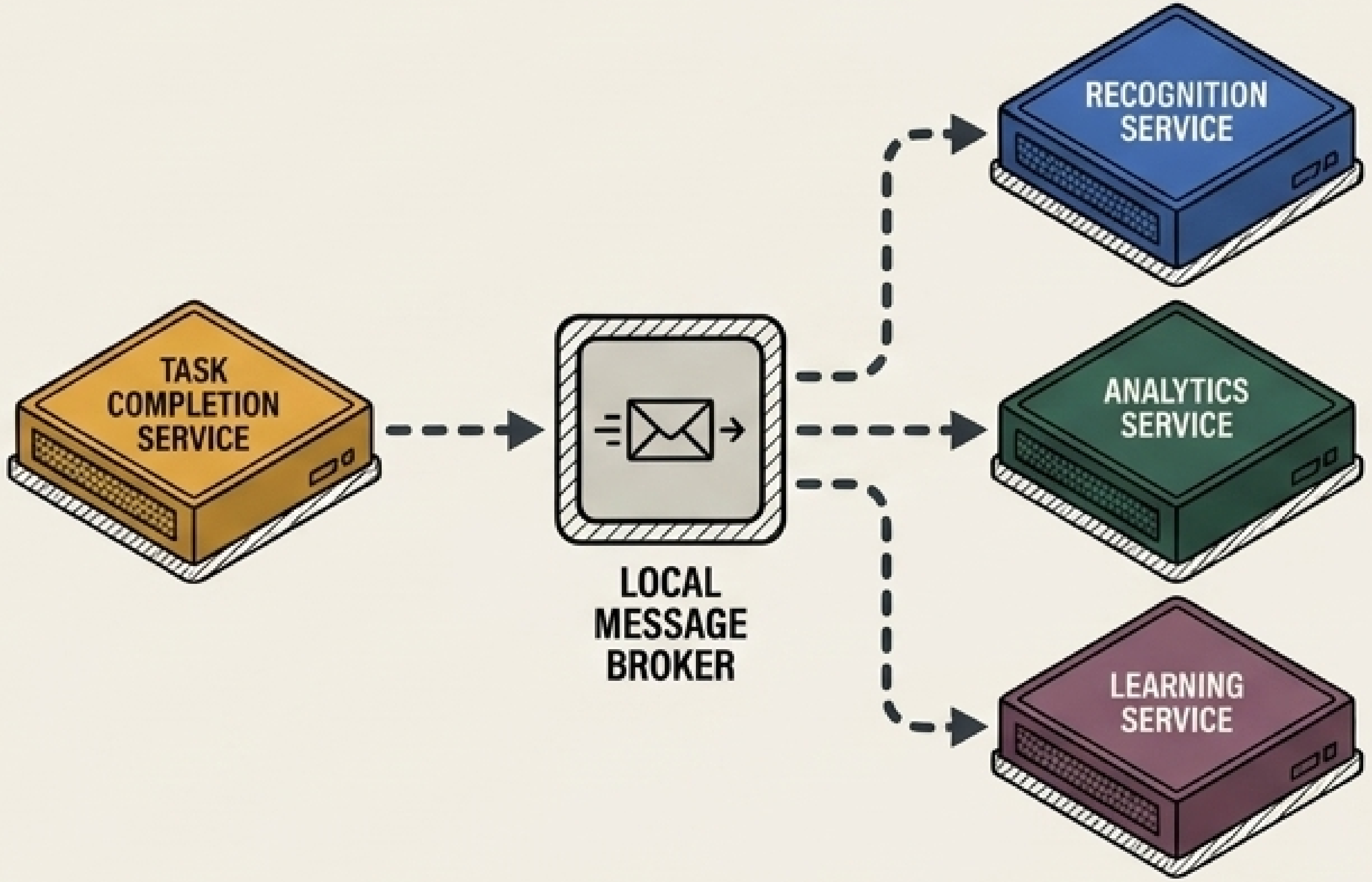
INDEPENDENT SERVICES COMMUNICATE THROUGH STRICT, VERSIONED CONTRACTS

SYNCHRONOUS (REST)



Used exclusively for immediate, required responses (e.g., the Project Service verifying user authorisation via the Admin Hub).

ASYNCHRONOUS (EVENTS)



Used for eventual consistency (e.g., Task Completion emitting a signal for downstream Recognition reactions).

ZERO-COST INFRASTRUCTURE CONSTRAINT: Messaging requires no paid cloud services. Any broker introduced must be free, open-source, and locally runnable.

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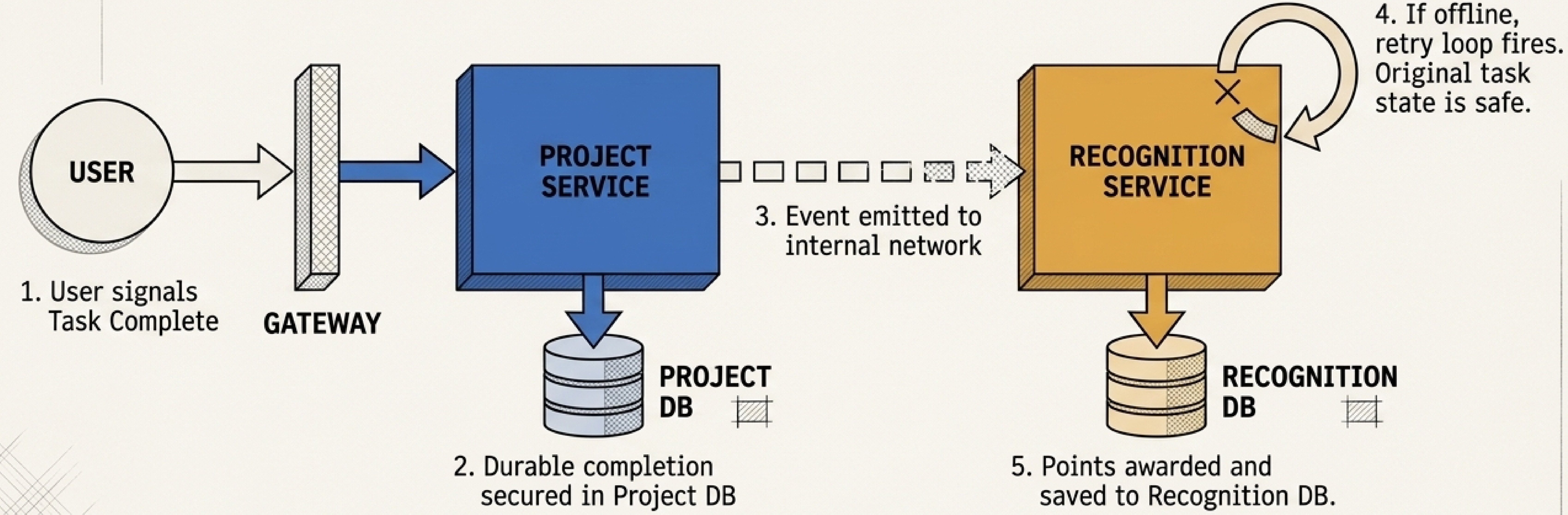
IDEMPOTENT WORKFLOWS GUARANTEE DATA INTEGRITY ACROSS BOUNDARIES



- **Durable State First:** The original task completion is secured before downstream triggers fire.
- **Idempotent Execution:** If an event is delivered twice due to network failure, the Recognition ledger ignores the duplicate request, ensuring balance accuracy.

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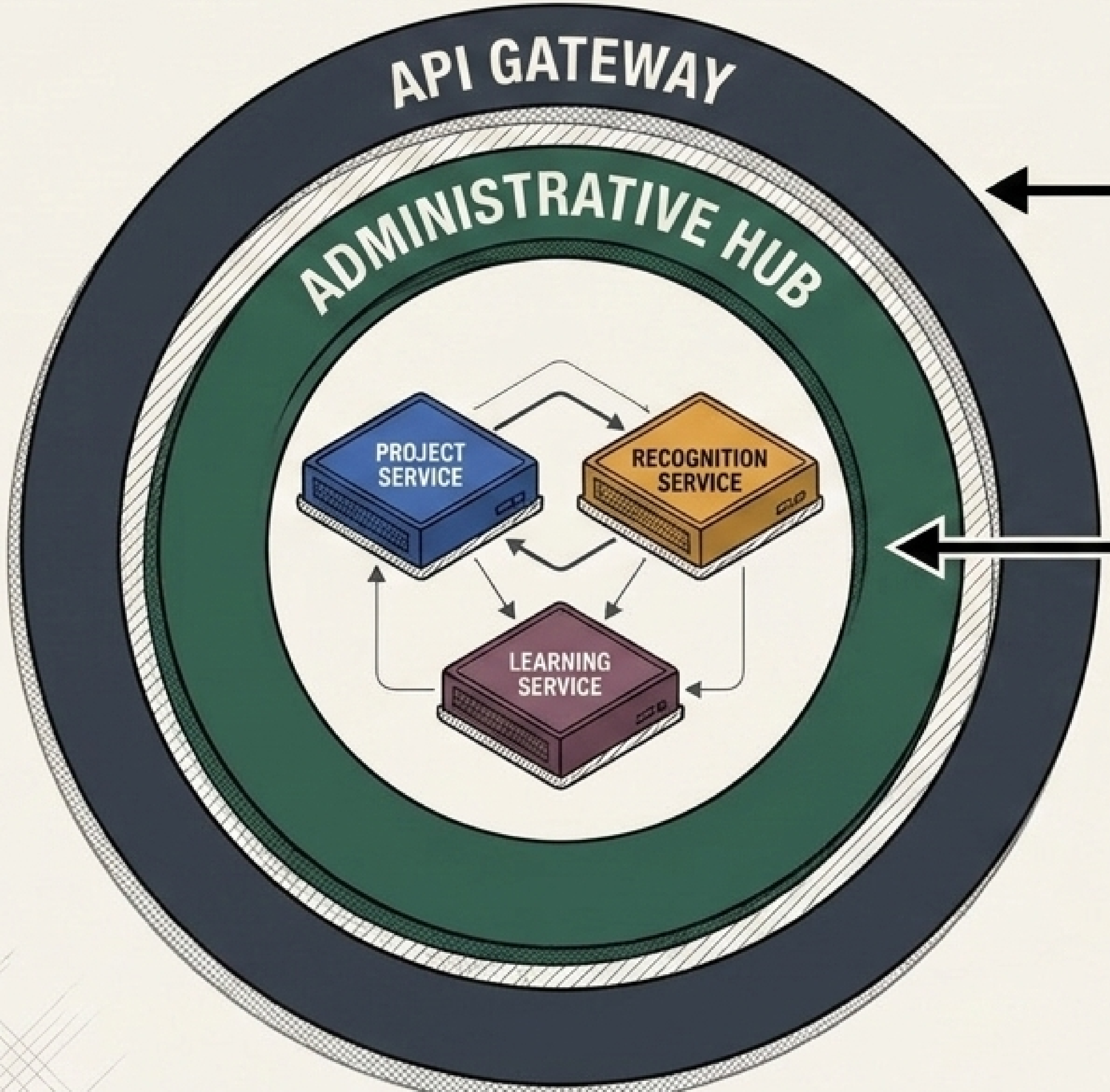
CENTRALISED IDENTITY ESTABLISHES A ZERO-TRUST INTER-SERVICE PERIMETER



Authentication (NFR-SEC-001): Verified at the security boundary.

Authorisation: Admin Hub defines privileges; individual services enforce them via internal trust.

Secret Management (NFR-SEC-002): Configuration is externalised. Secrets are never committed to the repository.

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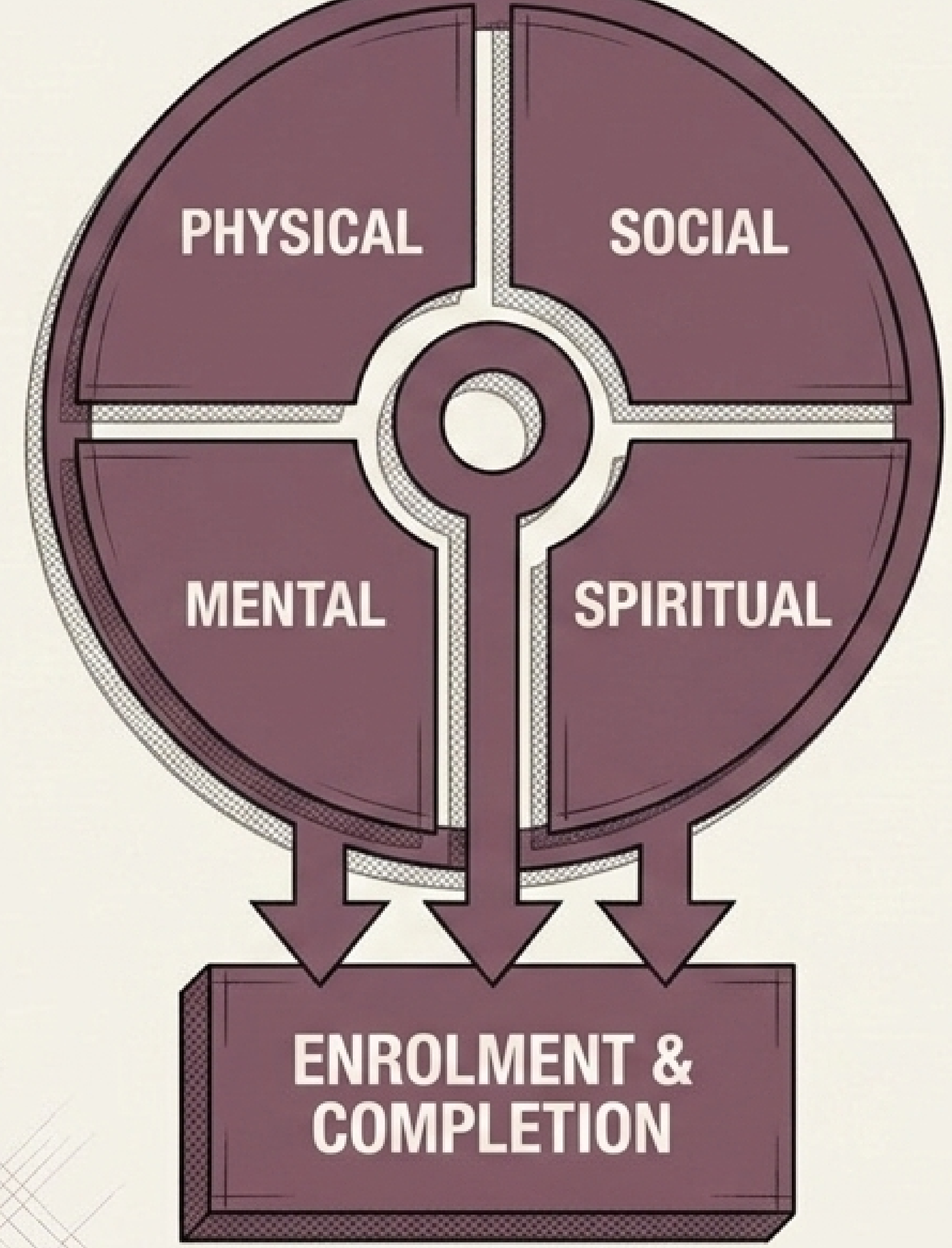
THE LEARNING FRAMEWORK CONFIGURES PROFESSIONAL DEVELOPMENT SAFELY



The Four Dimensions:
Configurable categories designed to track holistic professional milestones and competencies.

STRICT PROFESSIONAL BOUNDARY:

The system manages **professional development categories only.** It contains **absolutely zero medical, psychological, or religious advisory logic.**

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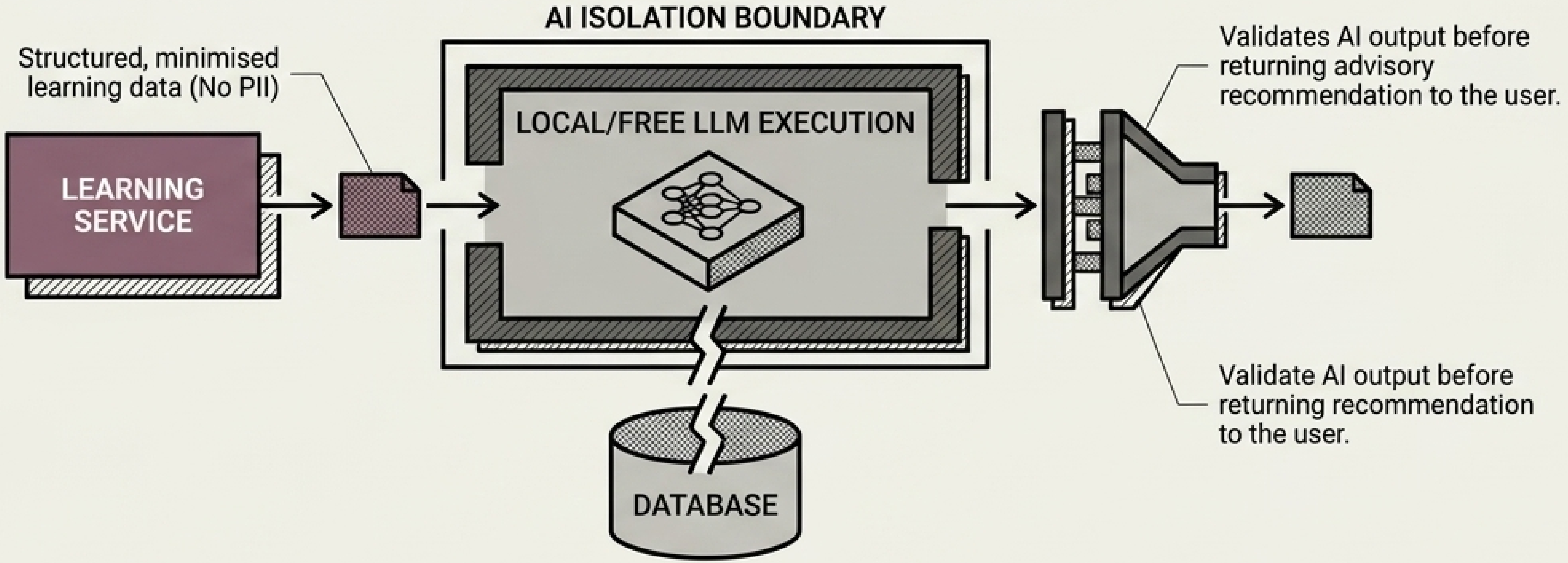
ARTIFICIAL INTELLIGENCE OPERATES EXCLUSIVELY WITHIN A STATELESS, ZERO-COST SANDBOX



NO CORE STATE OWNERSHIP:

AI is a powerful advisory tool, never the source of truth for core business data.

ZERO-COST PATH (NFR-COST-002):

AI models must have a free, locally runnable execution path for development and demonstration.

SAFE FALLBACK:

If the AI provider fails, core learning progression continues uninterrupted.

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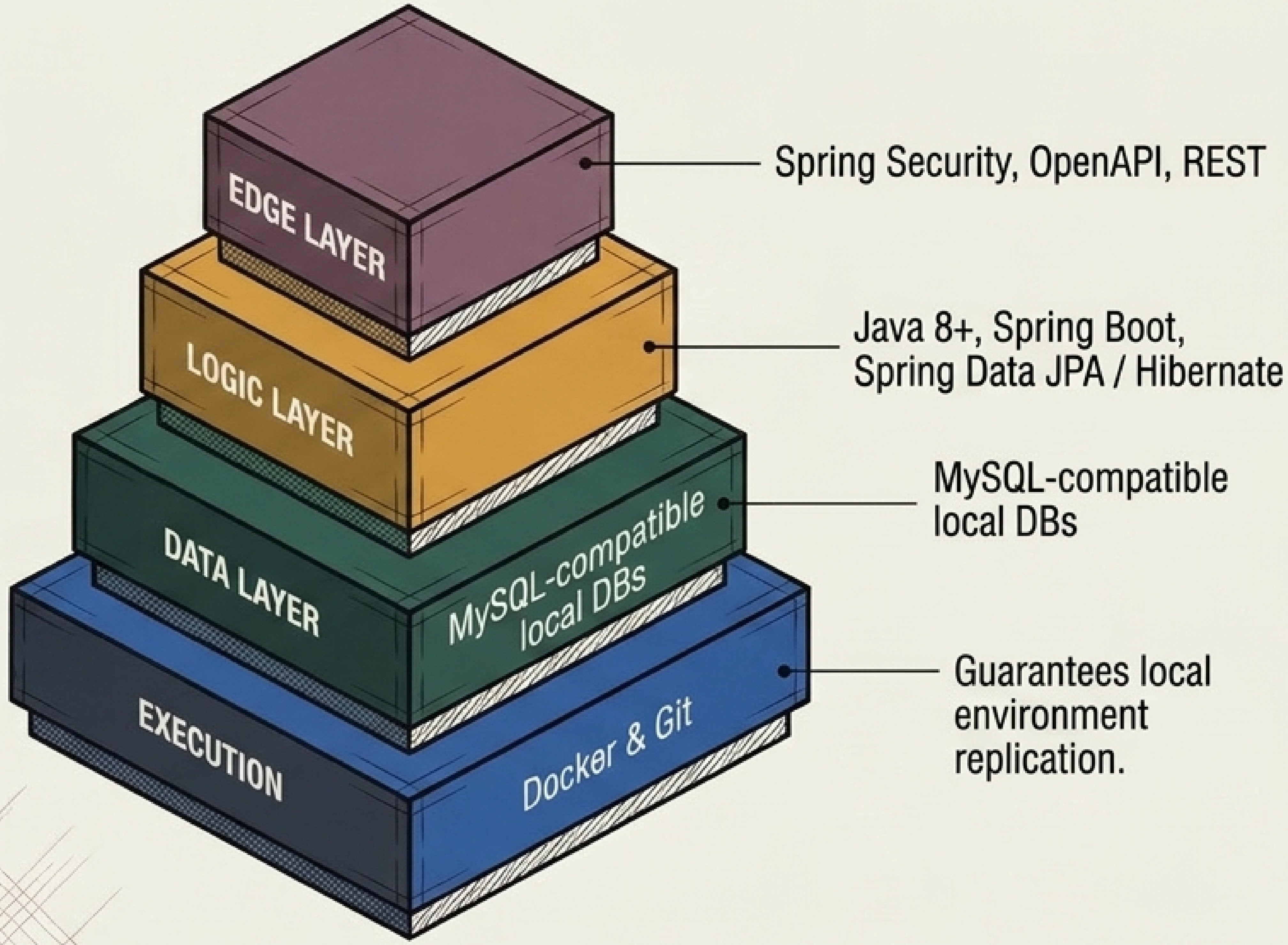
A RELIABLE TECHNOLOGY STACK ENFORCES THE ZERO-COST MANDATE



NFR-COST-001: Core development, testing, and demonstration require zero paid subscriptions or paid APIs.

Maintainability over Trends: The stack relies on proven, open-source Java technologies to ensure any engineer can onboard swiftly without proprietary vendor lock-in.

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THE ARCHITECTURE IS DESIGNED TO SURVIVE DOWNSTREAM FAILURE GRACEFULLY

Microservices amplify operational complexity. OmniGrowth counters this by expecting failure as a default state.

FAILURE SCENARIOS VS. SYSTEM RESPONSE

SCENARIO	RESPONSE
Invalid or Malformed Input	Immediate validation error at the gateway; no load on business logic.
Downstream API Offline	Controlled error and retry loop triggered; originating data remains fully intact in the source database.
Duplicate Event Received	Idempotent processing ignores the duplicate; ledgers remain accurate.
AI Component Unavailable	System executes a safe fallback; human-driven workflows proceed without disruption.

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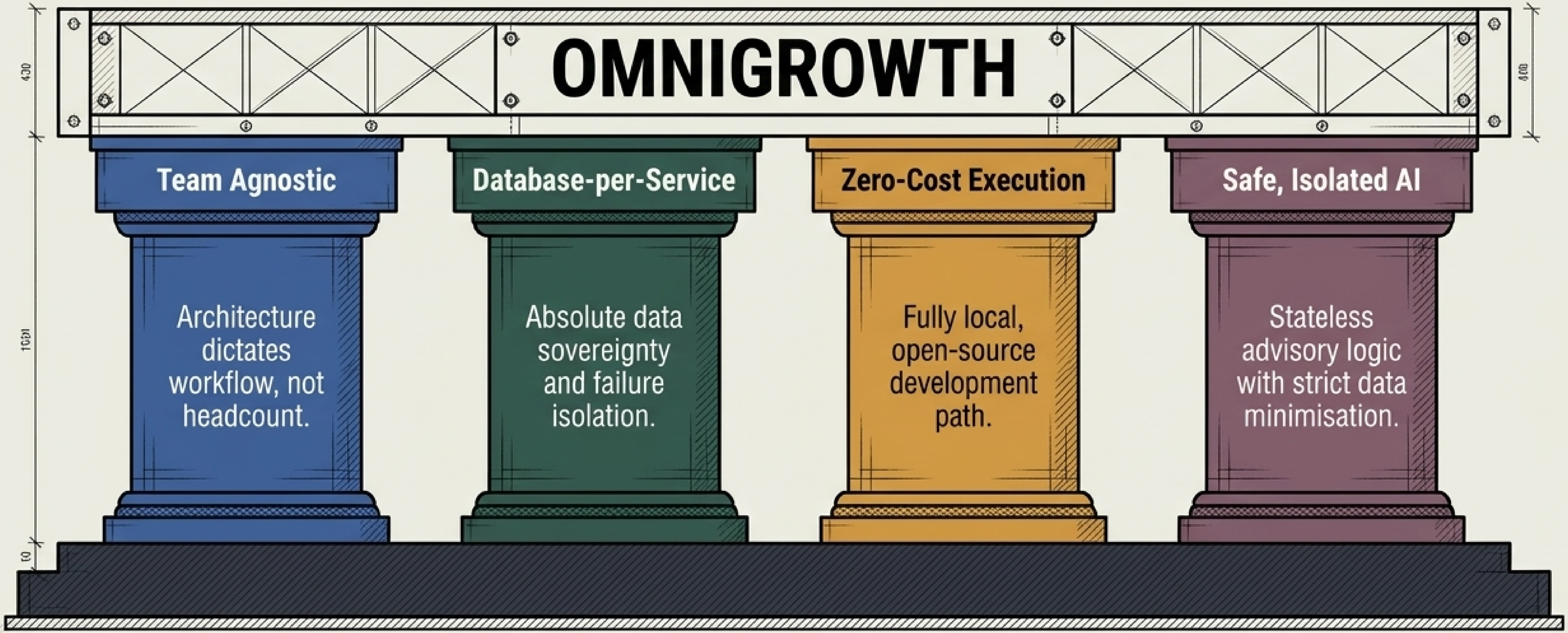
The OmniGrowth standard is a rigid contract for scalable engineering



These NFRs are non-negotiable. They ensure that whether built by one developer today or ten developers tomorrow, the system remains resilient, secure, and infinitely scalable.

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